

SALES FOOTBALL ANALYTICS ASSIGNMENT - POWERBI

PDF INFO: This is my tracking pdf where I define the pipeline for this assignment

1. Objective

Build a clean analytical model combining **Sales**, **Footfall**, and **Store Master** data to analyze:

- Total Sales
- Total Footfall
- Sales vs Footfall trend
- Store/Region performance
- Current year vs previous year comparison
- Data quality indicators

2. Dataset Overview (3 Datasets)

- a. Sales_data.csv : contains transaction-level sales data with store and date details.
- b. Footfall_data.xlsx : Daily store-level customer traffic counts.
- c. Store_master.csv : Master mapping of Store_ID to Store_Name and Region.

3. Issues Found (Data Quality Checks)

- a. Sales_data
 - Sales dates were in **multiple formats**, causing parsing issues.
 - **Sales_Amount had missing values** (nulls).
 - Some transactions had **Sales_Amount = 0**.
 - **Transaction_ID was not unique**, so it could not be treated as a primary key.
 - Store names had **trailing spaces**, creating inconsistent store labels.
- b. Footfall_data
 - Date was stored as **text** instead of Date format.
- c. Store_master
 - Store names had minor formatting inconsistencies (trailing spaces).

4. Assumptions Made

- Store_master is treated as the **source of truth** for Store_ID, Store_Name, and Region.
- Transaction_ID is treated as a reference field only, since it is not unique.
- Missing Sales_Amount values are not imputed, since revenue values should not be artificially created.
- Footfall data is assumed to represent total daily customer visits per store.

5. Fixes Applied (Power Query Cleaning)

- **Common Fixes (all tables)**
 - Standardized Store_ID datatype to **Text** for consistent joins.
 - Standardized Date datatype to **Date**.
- **Sales_Data.csv Fixes**
 - Standardized mixed date formats into a single Date column.
 - Trimmed and cleaned Store_Name to remove whitespace issues.
 - Converted Sales_Amount to numeric and retained missing values as null.
 - Removed only exact duplicate rows (same values across all columns).
 - Created a validity flag for sales records (Valid vs Invalid).
- **Footfall_Data.xlsx Fixes**
 - Converted Date from text to Date type.
 - Ensured Footfall_Count is numeric.
- **Store_Master.csv Fixes**
 - Trimmed and cleaned Store_Name.
 - Ensured Store_ID is unique.

6. Analytical Model

- Dimension Tables
 - DimDate (calendar table)
 - DimStore (from Store_Master)
- Fact Tables
 - FactSales (from Sales_Data)
 - FactFootfall (from Footfall_Data)

- Joining Logic

- FactSales joins to DimStore by Store_ID
- FactFootfall joins to DimStore by Store_ID
- Both fact tables join to DimDate by Date

7. Data Quality Indicators Used in BI Dashboard

- Missing Sales_Amount count
- Zero Sales_Amount count
- Duplicate Transaction_ID count
- Invalid sales transaction count
- % of valid sales records